Ecological Adaptations

Adaptation is the trait that enables an organism survive and reproduce in the environment it finds itself. Such trait or Adaption makes the organism fitted to their environment.

An ecological adaptation is any morphological, physiological, or behavioral attribute or trait that an organism has, that enables it survive and reproduce in its habitat.

There are three main types of ecological adaptation based on how organisms express their genetic traits. These are structural, behavioral, and physiological adaptation.

1. Structural adaptation has to do with the traits of how an organism's body is, such as color, skin, and shape of the body.
2. behavioral adaptation has to do with what the organism does or how it acts
3. physiological adaptation has to with mechanisms found in organisms that allow them to undertake specific metabolic activities to live in their natural habitat

Plant Adaptations and Ecological Classification of Plants

Plants are adapted to different environmental conditions

A. Adaptation to Light

Plants are classified into two based on their adaptation to light

1. Heliophytes:
Theses are Plants adapted to high light intensity or bright light. These plants have high photosynthetic, metabolic and respiratory rates E.g. Sun plants.

2. Sciophytes:
These are Plants adapted to low light intensity. These plants have low photosynthetic, metabolic and respiratory rates. They grow well under shade E.g. ferns and several herbs growing on the forest floor under the canopy of trees and shrubs.

B. Adaptations to water

Plants adapted to growing in water, water-logged soil, wetlands or on substrata with an abundance of water are Aquatic Plants (or Hydrophytes).

Plants living in aquatic environment **face several challenges in the form of low oxygen content, lack of soil and nutrients and less sunlight.**

These aquatic plants are divided into the following five categories, depending on whether they come in contact with water, air and soil.

1. **Free-floating hydrophytes:** These plants float freely on the water. They are in constant contact with water and air, but not with soil. Examples: Eichhornia and Pistia. Lemna, Wolffia, Trapa, Azolla, Salvinia,
2. **Rooted floating hydrophytes:** The roots of these plants are buried deep in the ground, while their leaves and flowers float on the water. Soil, water, and air all come into contact with these plants. Victoria, *Potomogeton, Marsilea,* Nelumbo and Nymphaea are examples.

3. **Rooted submerged hydrophytes:** These plants are completely immersed in water, have roots in the soil and do not interact with air. Examples: Hydrilla and Vallisneria.

4. **Submerged floating hydrophytes:** These plants are completely immersed in water and do not interact with soil and air. Examples: Utricularia and Ceratophyllum.

5. **Amphibious hydrophytes:** These plants are both aquatic and terrestrial. They grow in shallow water. Examples: Sagittaria, Scirpus, Typha and Ranunculus. They are also called emergent hydrophyte

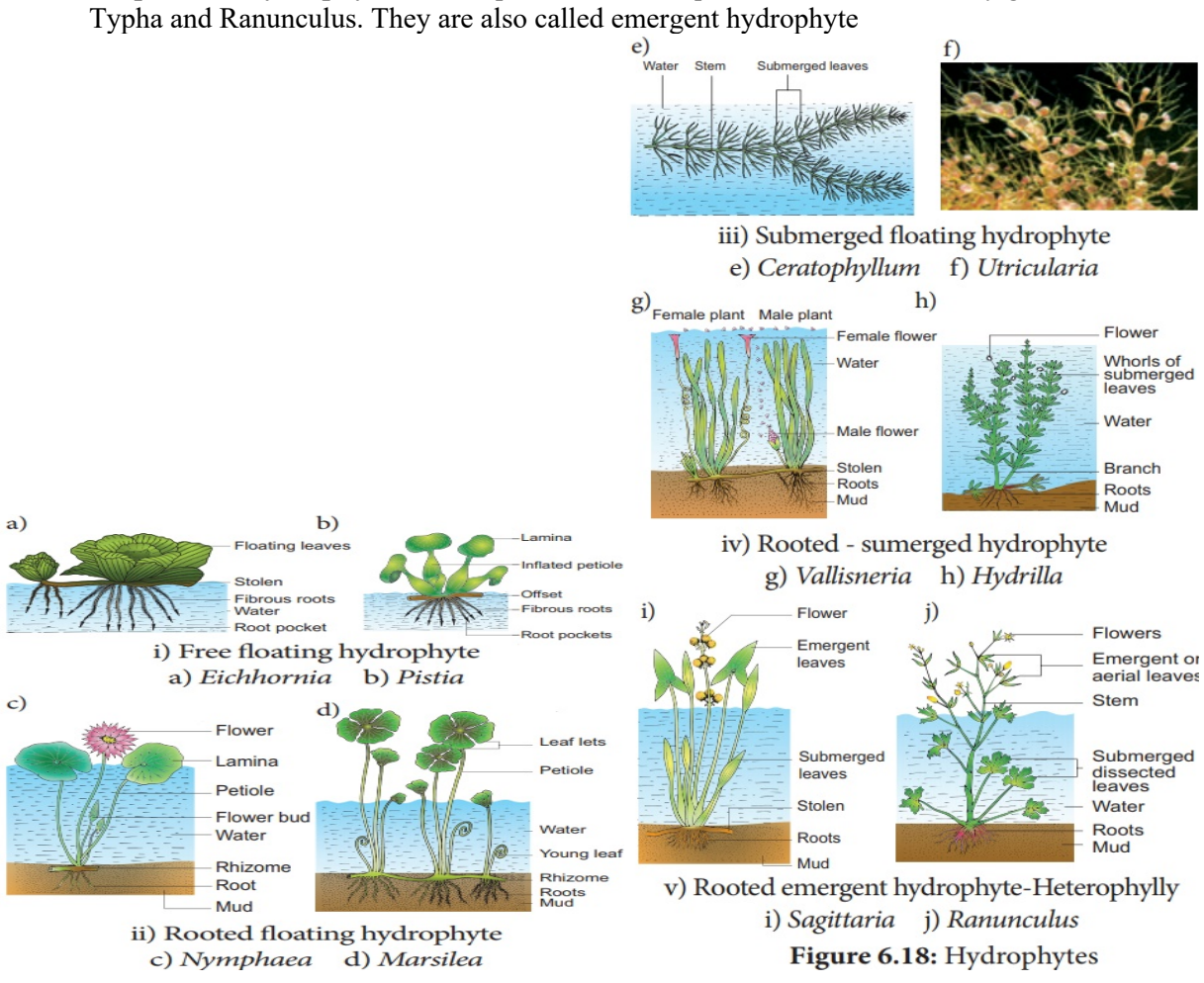


Figure 6.18: Hydrophytes

C. Adaptations to dry conditions

Plants adapted to growing in dry (or xeric conditions) or where the availability of water is negligible, are known as xerophytes.

There are different types of these plants

1. Ephemeral or Drought Escapers:
These plants complete their life cycle in a very short period only during rains. They live only for a period of 4-6 weeks, e.g., Tribulus terrestris, Solanum xanthocarpum.

2. Annuals or Drought Evaders:
The plants live for about 3 or 4 months, a few weeks longer than the rains. The size remains small, leaves have thick waxy, hairy coating with or without prickles to reduce transpiration, e.g., Echinops echinatus, Solanum surattense.

3. Succulents or Drought Resisting:
These xerophytes store water in their roots, stems, or leaves for drought periods. Succulents are stored in stems (Opuntia, Cacti), leaves (Aloe, Agave), and roots (Asparagus sp.).

4. Nonsucculents or Drought Enduring:
These are true xerophytes that actually endure drought conditions. They suffer from both external and internal dryness, e.g., Acacia, Ziziphus, Calotropis, etc.

5. Drought Avoiding Plants:
The life cycle of these plants is very short and these ephemerals complete their life cycle before summer comes, e.g., Argemone.

D. Adaptations to salty conditions

Plants adapted to growing in saline habitats are called halophytes.

The saline habitats have high concentrations of salts in soil or water. They occur in tidal marshes and coastal dunes, mangroves, along ocean edges and saline soils.

Examples of plants adapted to saline habitat are Rhizophora, Avicennia and Sundarban
The important adaptations of halophytes are as follows.

E. Adaptations of Plants to normal condition

Land plants adapted to grow under average conditions of temperature and moisture are called Mesophytes e.g Hibiscus

F. Adaptations of Plants to cold in Polar Regions

Polar Regions include one of the coldest regions on earth. And the vegetation found here is quite sparse.

Sedges, dwarf shrubs, grasses and certain mosses and lichens are some plant species found in these places.

The special features of these plants include short stature (not more than 12 inches), hairy parts, darker colour etc.

Only plants with shallow roots grow in the tundra because the deep layers of permanently frozen ice prevent the roots from breaking through the permafrost

Sometimes, trees hold on to dead leaves for insulation. However, larger plants such as trees as uncommon as the roots cannot penetrate deeply because of the permafrost.

G. Adaptations of Plants against Herbivory

Herbivory is the consumption of plant matter by any organism.

Since plants are the primary producers in an ecosystem, nearly all consumers depend on them for sustenance.

Therefore, to prevent herbivory, plants developed thorns, spines and chemicals.

Thorns are the most common form of deterrent.

Other plants might use chemicals to leave an unpleasant taste in the mouths of the herbivores. But some of these chemicals are very toxic and might result in the death if consumed.

Animal Adaptations

Like plants, animals also adapt to different environmental conditions to survive and flourish.

Animals have also developed a number of techniques in their bodies to meet the challenges of the environment in which they live.

Animals have adaptations to Particular feeding habits like carnivory and herbivory, Protection from predators, Camouflage for easy predation, Structural and behavioural adaptation to attract mate, Physiological and behavioural adaptation to environmental variations and stress conditions, e.g., migration, hibernation, aestivation, etc.

Some forms of animal adaptations are

1. Migration

It is a movement of an animal group to other places for food, climate, and other reasons.

Migration is of three types – daily, seasonal, and periodic.

The distance may be short or long. Long-distance migration is observed on Arctic Terns. They nest close to the North Pole in summer, fly south to Antarctica in autumn, and return North Pole again in each spring. Short-distance migration is observed in whales. Migration occurs in locusts when their number increases beyond the feeding capacity of the homeland.

2. Camouflage

Some animals have the capacity to blend with the surroundings or background. It is called camouflage.

It is a common adaptation in animals to remain unnoticed for protection or aggression, e.g., many insects, reptiles, and mammals.

It is difficult to distinguish leaf-like grasshoppers or Praying Mantis from the surrounding foliage. Similarly, stick insects, leaf insects and dead leaf butterflies can not be noticed unless and until they show movement.
Desert animals like lizards, camels, and lions are sand coloured and hardly visible against their background unless they move. The chameleon can change its colour according to its background. Camouflage is protective of animals that are preyed upon by others. It is advantageous in predation for predators like Preying Mantis as they remain unnoticed till the prey comes within their striking range.

3. Mimicry

The phenomenon of producing similarities with nonliving objects by which a living animal or plant species may conceal itself in the surroundings or obtain protection from enemies is known as mimicry.

Mimicry may be achieved either by altering the physical appearance or by changing behaviour. The living form which undergoes mimicry is known as mimetic and the object with which similarity is established is known as model.

Classification of Mimicry:
Mimicry may be categorized into three types namely, Protective mimicry, Aggressive mimicry, and Conscious mimicry.

1. Protective Mimicry:

To get protection from the predator animals, the mimicry shown by the animals is known as protective mimicry. This may again be of two types concealing mimicry and warning mimicry.
(a) Concealing mimicry:
Sometimes the animals try to conceal themselves in the surroundings so as to evade getting noticed by the enemies. In this case, the mimetic organism develops a similarity with an inanimate object either in physical appearance or in colour and because of this, it becomes difficult to locate the mimetic animal.

Examples:
Cryptolithode (a white-coloured crab) takes the shape of pebbles in the seashore and therefore they are difficult to be recognized.
A **type of butterfly named Papilio machaon** produces a pupa that when sticks to a tree branch appears green. Again when it attaches to the tree trunk or stone it becomes grey in colour.
The structure of the body and colour of the **butterfly Kallima paralekta** becomes such that it appears to be a dry leaf.
The **larva of the geometric moth, Selenia tetralunaria** when remains attached to the branch of the tree appears to be a twig.
The body shape and colour of the **insect Phyllium** appear like a fungus-infected green leaf.
The body structure of a stick insect is like that of a dry twig of a plant.
Arisaema is known as the Snake plant. In this plant, the exterior part of the inflorescence looks like a greenish-violet colour with the appearance of a cobra snake.

(b) Warning mimicry:
With the help of this mimicry, an animal may get protection from an enemy. In this case mimetic animal is usually non-poisonous, and this animal may take the shape of the model organism which is harmful or poisonous. In this case, both the body shape and colour of the organism may be altered. As a result of this, the enemies take the mimetic animal as harmful or poisonous.

Warning mimicry is of two types: i. Batesian and ii. Mullerian mimicry

Examples:
i. **Batesian mimicry** is when an unprotected and edible animal mimic another animal that is protected and unpalatable, the mimicry is termed as **Batesian mimicry**.

The **mimicry of butterfly Pieris** mimics tasteless Helicornius and gets protection from the predator animal.

ii. Mullerian mimicry is when two or more unpalatable species of animal mimic each other.

The **Ctenuchid moth** shows Mullerian mimicry by mimicking a type of wasp. In this case mimetic animals and models both get privilege.
Besides Batesian and Mullerian mimicry, there are certain examples of other warning mimicry.
Some non-poisonous coral snakes of the family Colubridae take the colour of the snakes of the poisonous Elaphidae family.
Some spiders live together with ants and take the colour ants.
Sesia crabroniformis, a type of moth mimics the dreaded wasp, Vespa crabro.

2. Aggressive mimicry:

This is the mimicry of the predator animals for capturing the prey.

The mimicry in this case helps the mimetic animal in attack instead of defence. The mimicry of this type is termed as aggressive mimicry.

Aggressive mimicry may be of two types such as i. Concealling mimicry and ii. Alluring mimicry.

i. **Concealing mimicry**: this is where the predator animal remains in ‘such a disguise that the prey cannot recognize it in the environment. As a result, the prey may easily be caught by the predator animal.
e.g., **some spiders** take the colour and shape of the flower when it remains within the flower. This helps the spider to conceal itself in the environment.

ii. Alluring mimicry: this is where the predator animal remains in such a disguise that the prey becomes attracted to it.
The **American horned toad Ceratophrys** sits quietly and rotates one finger over the head. Because of this the prey animals become attracted and go towards the toad. **The deep sea angler fish, Lophius** for capturing prey may exhibit alluring mimicry.

3. Conscious mimicry:

Some animals pretend as dead when they realize the possibility of any danger. Such mimicry is known as conscious mimicry. **American opossum, Didelphis**, stays like a dead animal when it senses any danger.

4. Colouration

In the living world, there is the existence of various colours. Like mimicry, animals may achieve adaptation through colouration also. However, in many cases, the animals may get protection through colouration.

Colouration may be divisible into two types namely concealing coloration and warning colouration.

i. **Concealing colouration**: With the help of such colouration the animals may conceal themselves within the surroundings in such a way that it becomes difficult to find them. If mimicry is associated with colouration, it gets special adaptation in nature.

ii. **Warning colouration**: The colouration, in this case, protects the animal in such a manner that the predator animal gets frightened to attack the prey. Batesian and Mullerian mimicry exhibit this type of mimicry. Dark spots on the wings of some moths confuse the predators and they get protection from this.

5. Echolocation

Bats are nocturnal flying mammals that do not employ eyesight for the location of their path, food, place of rest, etc.
They produce high-frequency sound which produces echoes after striking various objects on the principle of sonar.
Echoes are analyzed by the bats to know their path. Echolocation is also used by other animals, e.g., whales.

6. Adaptation to Excessive Cold (Cold Hardening)

Animals have also developed adaptations to extreme cold conditions.

Animals of colder areas **possess thick fur and sub-cutaneous fat**, they help in the conservation of heat.

During excessive cold, they may undergo hibernation. However, sea animals can not undergo **hibernation**.

Sessile animals cannot migrate. Certain sessile animals, such as barnacles and mollusks which live in very cold intertidal zones or northern shores, several insects and spiders resist the effect of cold by a process, known as **cold hardening**. Such animals are called **freeze-tolerant animals**. They possess ice-nucleating proteins, which induce ice formation in the extracellular spaces at very low sub-zero temperatures.

Some animals can tolerate environmental temperatures below 0°C by **accumulating glycerol or antifreeze proteins that lower the freezing point of their body fluids**. Such animals are called **freeze-avoiding animals.** The **fishes in the Antarctica region** remain active in cold seawater due to the presence of antifreeze compounds.

7. Adaptation to Water Scarcity

The **animals, which live in dry or xeric habitats**, are called **xerocoles**.

The xerocoles are of two types: i. drought evaders and ii. drought resistants.

i. **Drought evaders**: The drought-evading xerocoles adopt either a short annual life cycle or undergo aestivation or some other dormant state during the dry season.

The eggs and pupae of many insects (ants, bees, butterflies) and eggs of some invertebrates may remain dormant for several months during the dry season. In dry habitats, the birds breed and make nests during the rainy season, when there is abundant food.

ii. Drought resistants: These xerocoles can carry out their normal function throughout the year. They are able to resist aridity and heat through morphophysiological adaptations or by modifying their feeding activity patterns.

Some animals avoid heat by adopting nocturnal habits and remaining underground during the daytime.

Kangaroo and desert rats seldom drink water. They possess thick coats to minimize evaporative desiccation. 90% of their water requirement is met from metabolic water while 10% is obtained from food. Loss of water is minimized by producing nearly solid urine and feces.

Camel has a number of adjustments to desert conditions economical in water consumption, tolerance to fluctuations of temperature, and no sweating till body temperature rises to 55°-66°C. The animal produces dry feces and concentrated urine. During periods of non-availability of water the animal stores urea and does not produce urine. When water is available, a camel can rehydrate itself quickly by drinking a large quantity of water, some 80 liters in 10 minutes. Birds and reptiles in the desert retain water as their metabolic wastes are excreted in the form of an insoluble white compound called uric acid.

Cold-blooded animals entirely lack sweat glands as they rely on the external environment to regulate body temperature

8. Behavioural Adaptations in Desert Lizards

Desert lizards lack the physiological ability that mammals have to deal with high temperatures.
They keep their body temperature fairly constant by behavioural means.

They enjoy the sun and absorb heat when their body temperature drops below the comfort zone, but move into shade when the surrounding temperature starts increasing.

Some species are capable of burrowing into the soil to hide and escape from too much heat. Therefore, not only desert lizards but almost all reptiles are called cold-blooded animals, as they cannot maintain a constant inner temperature. So, behavioral adaptation helps them to do so.

Other Sources of Water
Natural sources of water, such as lakes and river are almost non-existent or are seasonal. So, animals derive water from desert plants such as cactus. Some insects also tap fluids such as saps and nectars from various parts of the plants.

Specialized Appendages
Animals like the jackrabbit have very large ears that have a network of blood vessels. When these animals rest in the shade, their enormous ears dissipate the excess heat from their bodies.

Mode of Nutrition in Plants and Animals

Mode of Nutrition in Plants

Plants primarily rely on autotrophic nutrition, specifically photoautotrophy. They use sunlight, water, and carbon dioxide to synthesize their own food through photosynthesis.

However, a few specialized plants exhibit heterotrophic or mixotrophic modes to supplement their energy needs.

1. Photoautotrophy (Primary Mode) The vast majority of plants are photoautotrophs.Mechanism: They use chlorophyll to capture light energy, converting it into chemical energy (sugars) via photosynthesis.Requirements: Sunlight, water, carbon dioxide, and essential soil minerals.

2. Heterotrophy (Specialized Exceptions) Some plants have lost their chlorophyll and rely on other organisms for nutrients.

Parasitic Plants: Extract nutrients directly from host plants using specialized roots called haustoria (e.g., Rafflesia or Mistletoe).

Mycoheterotrophic Plants: Do not photosynthesize; instead, they tap into mycorrhizal fungi to steal nutrients from photosynthetic trees (e.g., Indian Pipe).

3. Mixotrophy (Carnivorous Plants) Certain plants are autotrophic but supplement their diet with insects to survive in nutrient-poor soils.Mechanism: They perform photosynthesis but use modified leaves to trap and digest prey, absorbing vital nitrogen and phosphorus (e.g., Venus Flytrap, Pitcher Plants).

Mode of Nutrition in Animals

Animals rely on the heterotrophic mode of nutrition, meaning they cannot synthesize their own food and must depend on other organisms for sustenance. They obtain their nutrients through several distinct mechanisms. Animals exhibit a diverse range of nutritional modes based on how they obtain energy and carbon. Animals (metazoans) primarily rely on ingestive heterotrophy, while some have evolved fascinating symbiotic relationships.

Here is a breakdown of the primary nutritional modes found in animals:

1. Ingestion (Holozoic Nutrition) The vast majority of animals are holozoic, meaning they ingest (swallow) whole food or large organic particles into an internal digestive tract, where it is broken down internally. This mode is categorized by what these animals eat:

Herbivores: Animals that consume plants or algae (e.g., cows, rabbits, caterpillars).

Carnivores: Animals that consume other animals (e.g., lions, eagles, sharks).

Omnivores: Animals that consume both plant and animal matter (e.g., bears, humans, pigs).

Detritivores: Animals that feed on dead, decomposing organic matter and feces (e.g., earthworms, dung beetles).

2. Absorption (Osmotrophy) While common in fungi and bacteria, some highly specialized multicellular animals rely on absorbing dissolved organic nutrients directly from their environment.

Parasitic Worms: Endoparasites like tapeworms lack a digestive system (gut). Instead, they absorb predigested nutrients directly through their body walls from the host's intestine.

3. Filter/Suspension Feeding A sub-category of ingestion, this mode involves straining suspended food particles (like plankton or organic detritus) from water. Many aquatic animals draw water into their bodies and trap food using specialized structures.Examples: Bivalves (clams, oysters), baleen whales, and sponges.

4. Symbiotic Autotrophy / Chemotrophy Some animals do not produce their own food but harbor photosynthetic or chemosynthetic microbes within their tissues to provide them with nutrients.

Chemoautotrophy (via bacteria): Giant tube worms found at deep-sea hydrothermal vents have no mouth or gut. They rely on symbiotic bacteria inside their bodies that use hydrogen sulfide from the vents to produce organic molecules.

Photoautotrophy (via algae): Corals and giant clams harbor symbiotic photosynthetic algae (zooxanthellae) within their cells. The animals provide shelter, and the algae provide oxygen and sugars produced via photosynthesis.